An Interactive Augmented Reality Imaging System for Minimally Invasive Orthopedic Surgery

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***Abstract*—In orthopedic surgery, it is important for physicians to completely understand the three-dimensional bone structure for several procedures. To achieve this goal, it is required to image the patient several times using C-arm scanner from different positions during the surgery. This procedure is time consuming and increase the x-ray dose given to both patient and physician. In this paper, we propose an augmented reality imaging system for minimally invasive orthopedic surgery. The system is based on mapping the x-ray image to the real object such that the number of x-ray shots during the surgery can be significantly reduced. We consider two imaging scenarios that can fit with different cases. Results obtained through clinical data indicate that the proposed approach has a potential usefulness in real applications.**

Keywords-component; augmented reality; X-ray imaging; C-arm scanner; image registration

I. INTRODUCTION

Innovations in interventional surgery have created a strong demand for highly integrated solutions capable of supporting smooth workflows for increasingly complex procedures. Radiation exposure during the use of the C-arm scanner procedure is a continuing concern. Excellent image quality and lowest possible dose are vital requirements for patient and staff. The field of interventional surgery is growing. It is rapidly becoming an integral part of the patient treatment scheme in more and more clinical situations, helping to fight the most threatening diseases, prolonging patients' lives and improving their quality of life. From routine interventional procedures to new innovative endovascular and percutaneous treatments - efficient imaging techniques are a necessary base for them all. Minimally invasive procedures are also becoming more and more commonplace in orthopedic surgery. This is partly due to cost pressures and time constraints, but primarily to minimize tissue damage and improve patient recovery time. Advanced imaging tools offer surgeons ideal support for advanced cases. 3D imaging provides excellent support for delicate placement tasks, reduces the rate of second interventions, and has revolutionized the entire workflow.

Mobile C-arm is a useful imaging tool for image-guided interventional radiology. C-arm systems are provided with image intensifier (XRII) or flat-panel detectors. Essentially, C-arm CT systems requires scanners with flat-panel detectors for its ability to provide homogenous image quality and improve the resolution of low-contrast subjects compared to those equipped with image intensifier (Strobel, *et al*., 2009; Daly, *et al.*, 2008). However, C- arm systems equipped with XRIIs are widely used in several interventional procedures. Such systems can provide a high quality two-dimensional (2D) fluoroscopic images that facilitates minimal invasive surgery. However, it is likely that a physician is required to obtain several shots to be able to understand the internal structure and map it with visual image. This task is time consuming and requires a special skill obtained through training and experience. Orthopedic surgery is an example of common application performed through x-ray C-arm system. In many cases, it is difficult to observe the exact bone structures using 2D images acquired using conventional C-arm scanner in emergency situations. It requires several attempts to image the patient from different positions and physician requires a mental process to imagine how the 3D structure looks like before starting the interventional procedure. This process may be repeated several times during single procedure to follow up internal fixation using plates and screws, which cause a significant increase of radiation dose given to both patient and surgeon (Rampersaud *et al.*, 2000). The development of systems that maps the optical images with x-ray images was studied for long time. Several approaches are presented in this direction. The overlay accuracy and calibration is studied and improved by the work of (Chen *et al.*, 2013). In (Wang et al., 2016) propose an inverse approach to map x-ray images obtained from the C-arm using RGB camera with depth information.

This work aims to develop an augmented reality system to be used in interventional procedure to facilitate quick and low-dose orthopedic surgery. We consider two approaches, the first one output a fusion image obtained from x-ray C-arm scanner with true image obtained from a digital camera. The fused image is shown on a large display inside the surgery room. The second approach aims at projecting the x-ray image on the patient body, which provide an accurate and real matching on internal bone structures. Both approaches are developed through image registration and

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normalization process. We have conducted experiment using real data obtained from clinical C-arm scanner and a fixed digital camera. Results indicate an interesting fusion image that may reduce the requirement of several imaging during the orthopedic surgery. Interventional orthopedic surgery with interactive imaging system is a challenging field, but it is rewarding because the goal is to improve the health and the quality of human life.

II. Material and Methods

A. System Overview

The proposed system maps the x-ray image and real visual image in a way to reduce the need of several x-ray shots during surgery. First, an in-house made gantry control unit (GCU) is developed to control the scanner orbital rotation (Moataz *et al.*, 2015; Rashed *et al.*, 2015). This device is used to automatically control the motion of the C-shape arm of the scanner. Second, the C-arm rotation is traced using inertial measurement unit (IMU) sensor attached to the scanner gantry. Geometry information obtained from the IMU sensor is used to define the gantry position in the 3D space and synchronized with x-ray detector measurements. The imaging system with additional C-arm control devices is shown in Fig. 1. A C-arm CT system equipped with a flat-panel detector provides homogenous image quality and simultaneously improve the resolution of low-contrast. These system can provide a high quality 2D fluoroscopic images that facilitates minimal invasive surgery. However, 2D imaging techniques cannot obtained and localized deepness of body structures contrariwise 3D imaging techniques.

Two imaging scenarios are considered for image mapping. The first scenario is based on merging information obtained from the digital camera and C-arm scanner to generate a fusion image through a large display in the operation room. This setup is useful in case it is difficult to project the real image obtained from the digital camera over the patient body due to surgery circumstances or difficulties related to patient skin. The C-arm movement is operated using the GCU with foot pedal or GUI on a PC. The position of the scanner C-shape in the 3D space is measured using the IMU sensor and geometric data are sent to the processing unit using Bluetooth connection. X-ray trigger is modified to operate the x-ray tube and digital camera simultaneously to obtain x-ray and real images. Both images are sent to the processing unit for registration, normalization and fusion. Finally, fused image is sent to the display screen for physician assistant. The outline of this setup is shown in Fig. 2 (a). The second imaging scenario is configured by replacing the digital camera with an optical projector to directly project the x-ray image onto the patient body. Unlike the first scenario, only the x-ray image is available but there is no real image.

The system is calibrated such that the geometric information is used to register the x-ray image to be projected exactly over the patient body. Figure 2 (b) demonstrate the system setup configurations for the second imaging scenario.

Figure 1. The imaging system with GCU mounted to the C-arm (white arrow) and IMU sensor attached to the x-ray tube (red arrow). The system is tested with the Ziehm Vision R C-arm scanner installed in Suez Canal University Hospital, Ismailia, Egypt.

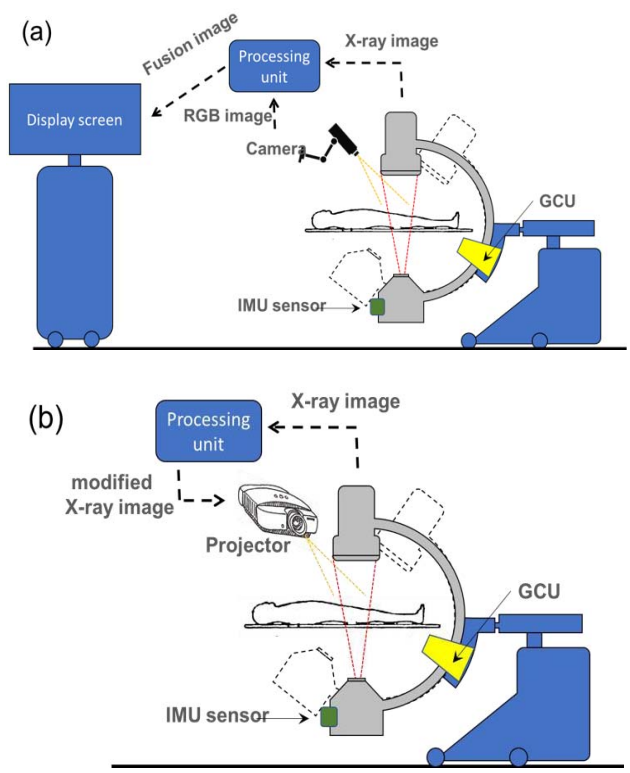

Figure 2. A sketch of the proposed system with two imaging scenarios. (a) A ceiling digital camera is used to acquire image of the scanned object to be merged with the x-ray image acquired using C-arm scanner. Image registration, normalization and merge are performed in the processing unit. (b) A ceiling optical projector is used to project the modified x-ray image into the patient body. The modified image is corrected using geometric information obtained from IMU sensor attached to the C-arm.

It is obvious that the second scenario may provide a more appropriate approach that facilitate the mapping between internal bone structures and real object during the surgery. However, in some cases, it is not possible to project x-ray image on the patient surface due to surgery requirements. To map the x-ray image with the real object we need to implement image processing techniques. For the first scenario, it is required to register two images obtained from different modalities. One is the color real image acquired using digital camera. The other is a gray-scale image measured by the C-arm scanner. For the second scenario, we map the x-ray image with real object. This requires system calibration to define several parameters such that the projected image is automatically mapped over the real object.

B. Image Acquisition and Registration

The system is operated by planning a C-arm movement using computer interface to be sent to the GCU. At each stopping point the system acquire x-ray image along with optical image. Both images are sent to the processing unit for image registration and fusion. Before the image registration, a preprocessing step is implement for view direction matching. The orientation of the virtual line connecting the central point of the detector and the x-ray source is computed using IMU sensor data. This virtual line is transferred to be aligned with the camera lens central line. This step is shown in Fig. 3.

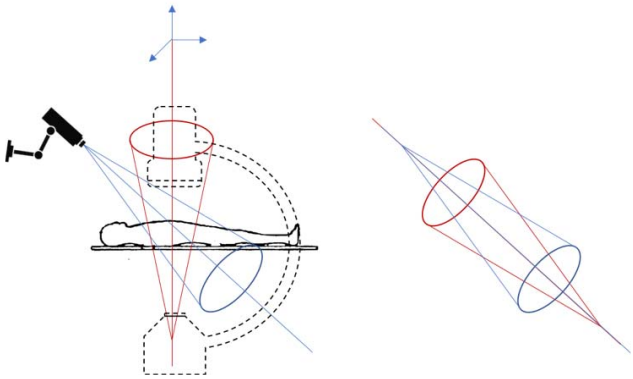

Figure 3. The sketch on the left demonstrate the geometries of the imaging system where there is a mismatch between the direction of x-ray acquisition view and optical image view. The x-ray image is transformed to fit with the direction view of optical image such that both views are matched as shown on the right-side sketch. A simple affine transformation is used in this preprocessing step.

The x-ray and optical images are registered by setting the optical image as fixed image and x-ray image as movable image. A non-rigid registration is used to fit the two images for the first imaging scenario. A deformable image registration method based on B-splines (Rueckert *et al.*, 1999). In this method, normalized mutual information is used as pixel-based similarity measure. This would lead to reduce the sensitivity of intensity value changes which is favor in multimodality image registration. The registration process is optimized using gradient decent method.

For the second scenario, we have used a control point registration technique. This method is based on setting up a metal markers on the scanning table. The optical and x-ray images are acquired for the markers with uniform background. An affine transformation parameters are computed using the coordinates of the control points. When the x-ray image is acquired during the imaging session, the same affine transform is applied to fit the image with the true object before sending the image to the projector.

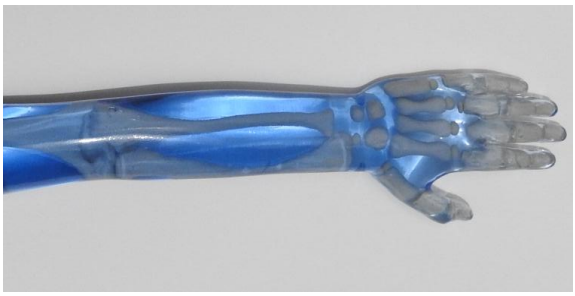

Figure 4. Left arm of CIRS pediatric anthropomorphic phantom (model 715). This phantom is used to mimic x-ray attenuation properties of a 5 years old child.

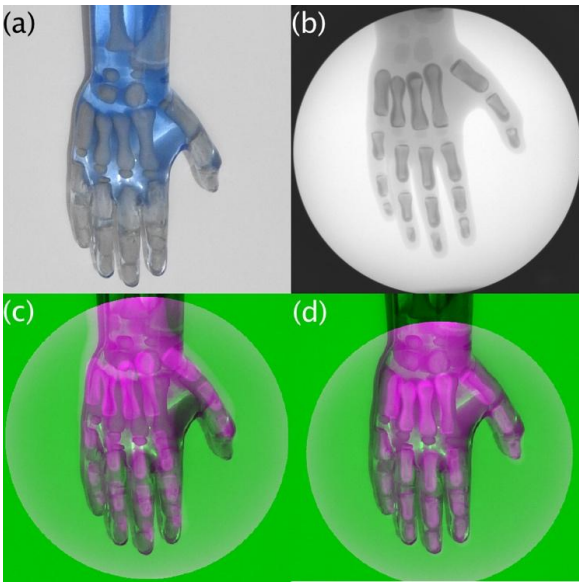

Figure 5. The optical image of the child hand phantom over a uniform white background obtained from the camera. (b) The x-ray image obtained from C-arm scanner. (c) A merge image using both images shown in (a) and (b). (d) A merge image using optical image (fixed) and x-ray image (movable).

III. Results and Discussion

The proposed system is evaluated using real experimental data obtained from clinical C-arm scanner shown in Fig. 1. The scanner C-arm is adjusted such that the center line between the x-ray source and the detector is almost aligned vertically to reduce the preprocessing step requirements. We have used CIRS pediatric anthropomorphic phantom (model 715) for this experiment. The part representing the left arm is shown in Fig. 4. The first imaging scenario shown in Fig. 2 (a) is used to image child hand phantom. A webcam is fixed to the room ceiling such that it can acquire the whole hand. Image acquired from the webcam is 1024×1024 pixels RGB

(JPEG formats) while the x-ray image is 512×512 pixels 8-bit gray scale (DICOM formats). The image registration algorithm described above in section II is used to register the two images. We consider the optical image as the fixed image and the x-ray image as the moving one. Finally, the two images are merged such that the bone structures are shown within the optical image. The images obtained from this experiment are shown in Fig. 5. In image shown in Fig. 5 (d), the anatomical bone structure shown inside the transparent phantom is almost matches the bone structures shown in the x-ray image.

For the second imaging scenario, we have used five metal rings as markers to compute the affine transformation for image matching. These metal markers are shown in Fig. 6 (a). It was inconvenient to project the x-ray image over the child hand phantom due to the transparency factor of the phantom. We have used real hand image of a volunteer healthy participant to test the second imaging scenario. First, we capture image with metal markers only to setup system parameters. Then the volunteer hand is added along with the markers. The x-ray image is transformed using affine transformation computed from the mapping of the marker's position data as a control points. The x-ray image is then inverted to better demonstrate the color of bones. Finally, the x-ray image is projected onto the hand volunteer. It worth mentioning that as the system is not moved, we can acquire several images without system calibration step using markers. If the C-arm is moved, the new 3D position can be acquired from the IMU sensor and the affine transformation is updated. Results are shown in Fig. 6.

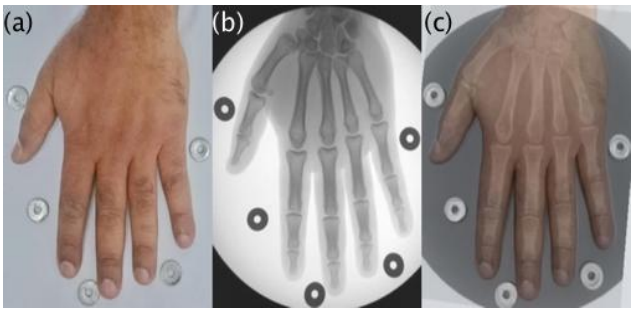

Figure 6. (a) A human hand with five metal markers. (b) x-ray image of the human hand with metal markers. (c) The moved modified x-ray image projected on the human hand.

IV. Conclusion

In this paper, we develop a new method for image fusion in interventional surgery. The developed method can provide a better environment for bone structure understanding inside the surgery room. Moreover, using the proposed system can significantly reduce the number of x-ray shots required during interventional procedures. The bone structure will be clear to the physician through a single x-ray shot. Results obtained from real data using child phantom and human hand indicate that the proposed method is expected to have a practical usefulness in clinical surgery.

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